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# =====
# Pixel Resolution vs Intensity Resolution
# Google Colab
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

image_name = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(image_name, cv2.IMREAD_GRAYSCALE)

# -----
# 1. Original Image
# -----

# -----
# 2. Pixel Resolution (Spatial Resolution)
# Reduce image size and enlarge again
# -----

small = cv2.resize(img,
                    (img.shape[1]//4, img.shape[0]//4),
                    interpolation=cv2.INTER_AREA)

pixel_resolution = cv2.resize(small,
                               (img.shape[1], img.shape[0]),
                               interpolation=cv2.INTER_NEAREST)

# -----
# 3. Intensity Resolution
# Reduce grayscale levels from 256 to 16
# -----

levels = 16

intensity_resolution = np.floor(img / (256/levels)) * (256/levels)
intensity_resolution = intensity_resolution.astype(np.uint8)

# -----
# Display Images
# -----

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plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(pixel_resolution, cmap='gray')
plt.title("Low Pixel Resolution")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(intensity_resolution, cmap='gray')
plt.title("Low Intensity Resolution")
plt.axis('off')

plt.tight_layout()
plt.show()

# -----
# Comparison
# -----

print("\nComparison Between Pixel Resolution and Intensity Resolution\n")

print("1. Pixel Resolution")
print("- Refers to the number of pixels in the image.")
print("- Lower pixel resolution reduces image sharpness.")
print("- Fine anatomical structures become blurry.")
print("- Important in detecting small tumors or fractures.")

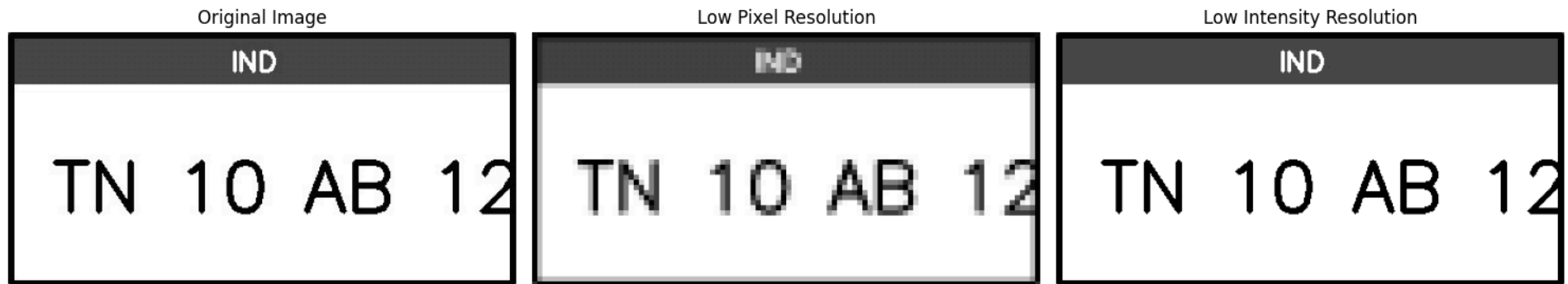
print("\n2. Intensity Resolution")
print("- Refers to the number of gray levels.")
print("- Lower intensity resolution causes banding.")
print("- Small contrast differences between tissues are lost.")
print("- Important in distinguishing soft tissues in MRI and CT scans.")

print("\nMedical Imaging Example")
print("- High Pixel Resolution helps identify tiny blood vessels, tumors, and fractures.")
print("- High Intensity Resolution helps differentiate healthy tissue from diseased tissue.")
```

Choose Files number_plate.jpg

number_plate.jpg(image/jpeg) - 14891 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg



Comparison Between Pixel Resolution and Intensity Resolution

1. Pixel Resolution

- Refers to the number of pixels in the image.
- Lower pixel resolution reduces image sharpness.
- Fine anatomical structures become blurry.
- Important in detecting small tumors or fractures.

2. Intensity Resolution

- Refers to the number of gray levels.
- Lower intensity resolution causes banding.
- Small contrast differences between tissues are lost.
- Important in distinguishing soft tissues in MRI and CT scans.

Medical Imaging Example

- High Pixel Resolution helps identify tiny blood vessels, tumors, and fractures.
- High Intensity Resolution helps differentiate healthy tissue from diseased tissue.